DOCKERFILE WORKDIR AND DOCKER NETWORK

DOCKERFILE WORKDIR:

The WORKDIR instruction in a Dockerfile sets the working directory for all subsequent instructions like RUN, CMD, ENTRYPOINT, COPY, and ADD. It helps streamline your Dockerfile, making it cleaner and less error-prone by avoiding the need to specify absolute paths repeatedly.

- using “vi Docker” command to edit the file using an workdir in this code instruction for an working directory using an run command of this code to using of linux command for nginx web server.

```
[root@ip-172-31-77-224 ec2-user]# ls
Dockerfile  index.html
[root@ip-172-31-77-224 ec2-user]# docker images
REPOSITORY   TAG       IMAGE ID       CREATED        SIZE
firstimage   v1        d43ffd5359c3   27 hours ago   296MB
[root@ip-172-31-77-224 ec2-user]# cat Dockerfile
FROM amazonlinux:2
WORKDIR /usr/share/nginx/html/
RUN yum update -y && \
    amazon-linux-extras install nginx1 -y && \
    yum clean all
COPY . .

# Start nginx in foreground
CMD ["nginx", "-g", "daemon off;"]
```

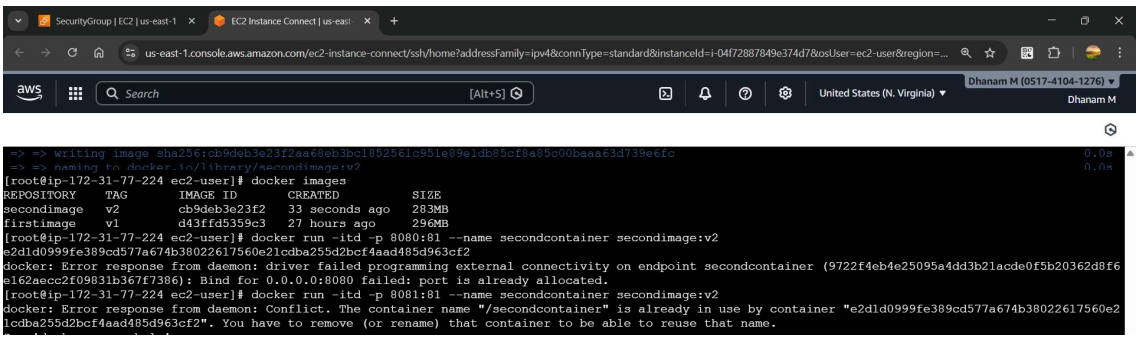
- using “docker run -itd -p <hostport>:<containerport> --name <container name:<containertag

```
[root@ip-172-31-77-224 ec2-user]# docker run -itd -p 8080:80 --name firstcontainer firstimage:v1
docker: Error response from daemon: Conflict. The container name "/firstcontainer" is already in use by container "2cae8b653a6dd4e12ec890a595234793ca4f27
eed56a1525b76f2eea0cacae17". You have to remove (or rename) that container to be able to reuse that name.
See 'docker run --help'.
[root@ip-172-31-77-224 ec2-user]# docker ps -a
CONTAINER ID   IMAGE           COMMAND                  CREATED        STATUS                     PORTS     NAMES
2cae8b653a6d   firstimage:v1   "nginx -g 'daemon of…"   27 hours ago   Exited (0) 26 hours ago            firstcontainer
[root@ip-172-31-77-224 ec2-user]# docker restart ^C
[root@ip-172-31-77-224 ec2-user]# docker restart 2cae8b653a6d
2cae8b653a6d
[root@ip-172-31-77-224 ec2-user]# docker ps
CONTAINER ID   IMAGE           COMMAND                  CREATED        STATUS         PORTS                                       NAMES
2cae8b653a6d   firstimage:v1   "nginx -g 'daemon of…"   27 hours ago   Up 8 seconds   0.0.0.0:8080->80/tcp, :::8080->80/tcp   firstcontainer
[root@ip-172-31-77-224 ec2-user]#
```

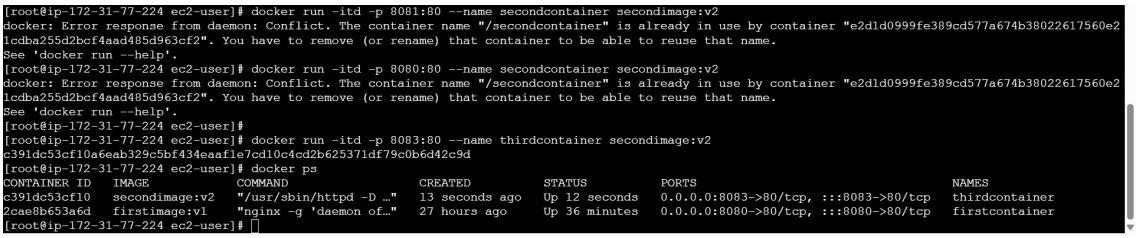
- using “docker ps” to list an running container “docker exec -it <container id> /bin/bash” command to inside their container “ls” to list the file “pwd” to display an absolute path of current file system.

```
[root@ip-172-31-77-224 ec2-user]# docker ps
CONTAINER ID   IMAGE           COMMAND                  CREATED        STATUS         PORTS                                       NAMES
2cae8b653a6d   firstimage:v1   "nginx -g 'daemon of…"   27 hours ago   Up 8 seconds   0.0.0.0:8080->80/tcp, :::8080->80/tcp   firstcontainer
[root@ip-172-31-77-224 ec2-user]# docker exec -it 2cae8b653a6d /bin/bash
bash-4.2#
bash-4.2# ls
404.html  50x.html  Dockerfile  icons  index.html  nginx-logo.png  poweredby.png
bash-4.2# pwd
/usr/share/nginx/html
bash-4.2#
```

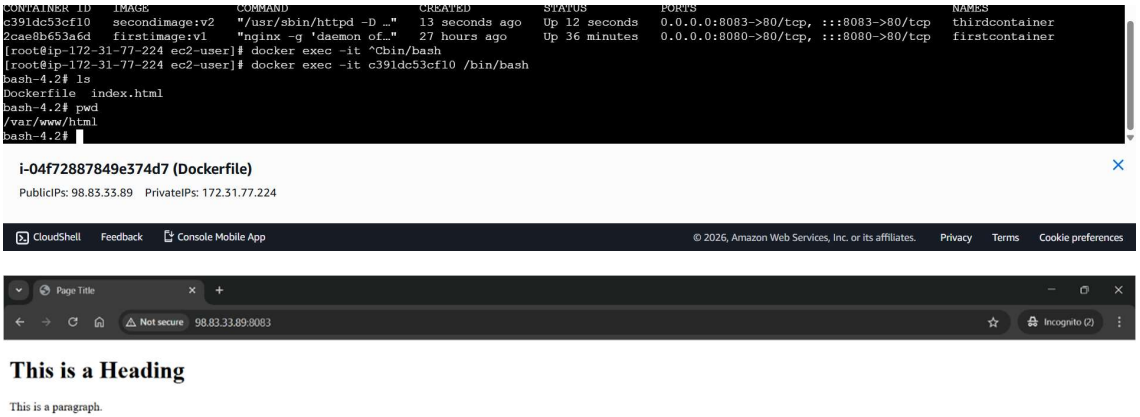
- using “vi Docker” command to edit the file using an workdir in this code instruction for an working directory using an run command of this code to using of linux command for apache web server.



- using “docker run -itd -p <hostport>:<containerport> --name <container name:<containertag



- using “docker ps” to list an running container “docker exec -it <container id> /bin/bash” command to inside their container “ls” to list the file “pwd” to display an absolute path of current file system.



DOCKER NETWORK:

- Docker networking is a system that allows Docker containers to communicate with each other, the host machine, and the outside world.
- Using “docker network” command to manage an docker network. “docker network create <network name> command to use create an network. “docker ls” command to list the docker command.

```
Usage:  docker network COMMAND

Manage networks

Commands:
  connect     Connect a container to a network
  create      Create a network
  disconnect  Disconnect a container from a network
  inspect     Display detailed information on one or more networks
  ls          List networks
  prune       Remove all unused networks
  rm          Remove one or more networks

Run 'docker network COMMAND --help' for more information on a command.
[root@ip-172-31-77-224 ec2-user]# docker network create mynetwork
3d18407aa9ef479ca249cce613c6754dd1fd05819669912029ac993d509ed42f
[root@ip-172-31-77-224 ec2-user]# ls
Dockerfile  index.html
[root@ip-172-31-77-224 ec2-user]# docker network ls
NETWORK ID     NAME      DRIVER    SCOPE
660a44d8b4c5   bridge    bridge    local
78c70391e742   host      host      local
3d18407aa9ef   mynetwork bridge    local
79fe2cec1e2b   none      null      local
[root@ip-172-31-77-224 ec2-user]#
```

i-04f72887849e374d7 (Dockerfile)

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- Using “docker connect network name <containerid>” command to use container to connect an network.

```
                "GlobalIPv6Address": "",
                "GlobalIPv6PrefixLen": 0,
                "DriverOpts": null,
                "DNSNames": null
            },
            "mynetwork": {
                "IPAMConfig": {},
                "Links": null,
                "Aliases": [
                    "c391dc53cf10"
                ],
                "MacAddress": "02:42:ac:12:00:02",
                "NetworkID": "3d18407aa9ef479ca249cce613c6754dd1fd05819669912029ac993d509ed42f",
                "EndpointID": "68be49e145e99e0db13aff7d04bfe29798fe6717bc6685c21f8882175972f69c",
                "Gateway": "172.18.0.1",
                "IPAddress": "172.18.0.2",
                "IPPrefixLen": 16,
                "IPv6Gateway": "",
                "GlobalIPv6Address": "",
                "GlobalIPv6PrefixLen": 0,
                "DriverOpts": {},
                "DNSNames": [
                    "thirdcontainer",
                    "c391dc53cf10"
                ]
            }
        }
    }
]
```

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- Using “docker run -itd -p 8085:80 –name netw –network mynetwork <imagename:imageid>” command to use run a image an netw container of an my network.

```
78c7039fe742   host        host      local
3d18407aa9ef   mynetwork   bridge    local
79fe2cec1e2b   none        null      local
[root@ip-172-31-77-224 ec2-user]# docker images
REPOSITORY    TAG       IMAGE ID       CREATED          SIZE
secondimage   v2        cb9deb3e23f2   24 minutes ago   283MB
firstimage    v1        d43ffd5359c3   28 hours ago     296MB
[root@ip-172-31-77-224 ec2-user]# docker run -itd -p 8084:80 --name netw --network mynetwork secondimage:v3
Unable to find image 'secondimage:v3' locally
docker: Error response from daemon: pull access denied for secondimage, repository does not exist or may require 'docker login': denied: requested access
 to the resource is denied.
See 'docker run --help'.
[root@ip-172-31-77-224 ec2-user]# docker run -itd -p 8084:80 --name netw --network mynetwork thirdimage:v3
Unable to find image 'thirdimage:v3' locally
docker: Error response from daemon: pull access denied for thirdimage, repository does not exist or may require 'docker login': denied: requested access
to the resource is denied.
See 'docker run --help'.
[root@ip-172-31-77-224 ec2-user]# docker run -itd -p 8084:80 --name netw --network mynetwork secondimage:latest
Unable to find image 'secondimage:latest' locally
docker: Error response from daemon: pull access denied for secondimage, repository does not exist or may require 'docker login': denied: requested access
 to the resource is denied.
See 'docker run --help'.
[root@ip-172-31-77-224 ec2-user]# docker run -itd -p 8084:80 --name netw --network mynetwork secondimage:v2
54fdd568eadd507ab66f3afcdb43691685cc286a8e3ffc2351fc803d79c69e09
[root@ip-172-31-77-224 ec2-user]# docker ps
CONTAINER ID   IMAGE             COMMAND                  CREATED          STATUS              PORTS                                          NAMES
54fdd568eadd   secondimage:v2    "/usr/sbin/httpd -D …"   32 seconds ago   Up 31 seconds       0.0.0.0:8084->80/tcp, :::8084->80/tcp          netw
c391dc53cf10   secondimage:v2    "/usr/sbin/httpd -D …"   25 minutes ago   Up 25 minutes       0.0.0.0:8083->80/tcp, :::8083->80/tcp          thirdcontainer
2cae8b653a6d   firstimage:v1     "nginx -g 'daemon of…"   28 hours ago     Up About an hour    0.0.0.0:8080->80/tcp, :::8080->80/tcp          firstcontainer
[root@ip-172-31-77-224 ec2-user]#
```

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```
            "GlobalIPv6PrefixLen": 0,
            "IPAddress": "",
            "IPPrefixLen": 0,
            "IPv6Gateway": "",
            "MacAddress": "",
            "Networks": {
                "mynetwork": {
                    "IPAMConfig": null,
                    "Links": null,
                    "Aliases": [
                        "54fdd568eadd"
                    ],
                    "MacAddress": "02:42:ac:12:00:04",
                    "NetworkID": "3d18407aa9ef479ca249cce613c6754dd1fd05819669912029ac993d509ed42f",
                    "EndpointID": "8e6c6c0e1f79280ff76ab72dc6c53cb92fc3dd3f2b61184f6f0f45d20a7d52be",
                    "Gateway": "172.18.0.1",
                    "IPAddress": "172.18.0.4",
                    "IPPrefixLen": 16,
                    "IPv6Gateway": "",
                    "GlobalIPv6Address": "",
                    "GlobalIPv6PrefixLen": 0,
                    "DriverOpts": null,
                    "DNSNames": [
                        "netw",
                        "54fdd568eadd"
                    ]
                }
            }
        }
    }
```


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